Spații vectoriale
- probleme propuse -

(1.) Se consideră vectorii $u_1 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$; $u_2 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$; $u_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \in \mathbb{R}^3$.

Fie $U = Sp\{u_1, u_2, u_3\}$.

a) Demonstrați că vectorul $u = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$ aparține subspațiului U.

b) Determinați o bază în U și coordonatele vectorului u în raport cu această bază.

Soluție:

a) Se observă că $u = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = u_1 - u_3 \Rightarrow$

$\Rightarrow u \in Sp\{u_1, u_2, u_3\} = U$.

b) $U = Sp\{u_1, u_2, u_3\} \Rightarrow \{u_1, u_2, u_3\}$ SG pt U (1)

$\operatorname{rang}(u_1 | u_2 | u_3) = \operatorname{rang}\begin{pmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{pmatrix} = 2$, deoarece

$\begin{vmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 3 & 1 & -1 \end{vmatrix} = -3 + 0 + 3 - (0 + 1 - 1) = 0$ și $\begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} = 3 - 1 = 2$

$\Rightarrow \{u_1, u_2, u_3\}$ SLD, dar $\{u_1, u_2\}$ SLI (2)

$\overset{(1)(2)}{\Rightarrow} \{u_1, u_2\}$ bază pt U.

Observăm că $u_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{2}\left[\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}\right] = \frac{1}{2}(u_2 - u_1)$

$\Rightarrow u = u_1 - u_3 = u_1 - \frac{1}{2}u_2 + \frac{1}{2}u_1 = \frac{3}{2}u_1 - \frac{1}{2}u_2$

$\Rightarrow$ coord lui u în raport cu baza $\{u_1, u_2\}$ sunt $\begin{pmatrix} 3/2 \\ -1/2 \end{pmatrix}$.

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(2) Fie polinoamele $P_1[X] = 1 + 2X + X^3$; $P_2[X] = 2 + X + 2X^2$; $P_3[X] = 3 + X^2 + X^3$.

a) Demonstrați că P_1, P_2, P_3 sunt liniar independente. Determinați o bază în $\mathbb{R}_3[X]$ care să conțină cele trei polinoame de mai sus.

b) Aflați coordonatele polinomului $P[X] = 3X + X^2$ în raport cu baza determinată.

Soluție:

a) $\{P_1, P_2, P_3\}$ SLi $\Leftrightarrow$ pt $a, b, c \in \mathbb{R}$ avem

$aP_1 + bP_2 + cP_3 = 0 \Rightarrow a = b = c = 0$

$\Leftrightarrow a(1 + 2X + X^3) + b(2 + X + 2X^2) + c(3 + X^2 + X^3) = 0 \Leftrightarrow$

$\Leftrightarrow (a + 2b + 3c) + (2a + b)X + (2b + c)X^2 + (a + c)X^3 = 0 \Leftrightarrow$

$$\Leftrightarrow \begin{cases} a + 2b + 3c = 0 \\ 2a + b = 0 \mid \cdot(2) \Rightarrow 4a + 2b = 0 \\ 2b + c = 0 \\ a + c = 0 \end{cases} \Rightarrow \begin{cases} \ominus \Rightarrow 5a = 0 \Rightarrow a = 0 \Rightarrow \\ \Rightarrow c = 0 \Rightarrow b = 0 \\ \text{(Adevărat)} \end{cases}$$

$\Rightarrow \{P_1, P_2, P_3\}$ SLi.

$\dim_{\mathbb{R}} \mathbb{R}_3[X] = 4 \Rightarrow$ pt a obține o bază, mai este nevoie de un polinom $P_4[X]$

Alegem $P_4[X] = 1$.

$\{P_1, P_2, P_3, P_4\}$ SLi $\Leftrightarrow$ pt $\alpha, \beta, \gamma, \delta \in \mathbb{R}$ avem

$\alpha P_1 + \beta P_2 + \gamma P_3 + \delta P_4 = 0 \Rightarrow \alpha = \beta = \gamma = \delta = 0.$

$\Leftrightarrow \alpha(1 + 2X + X^3) + \beta(2 + X + 2X^2) + \gamma(3 + X^2 + X^3) + \delta \cdot 1 = 0 \Leftrightarrow$

$\Leftrightarrow (\alpha + 2\beta + 3\gamma + \delta) + (2\alpha + \beta)X + (2\beta + \gamma)X^2 + (\alpha + \gamma)X^3 = 0 \Leftrightarrow$

$$\Leftrightarrow \begin{cases} \alpha + 2\beta + 3\gamma + \delta = 0 \\ 2\alpha + \beta = 0 \mid \cdot(2) \Rightarrow 4\alpha + 2\beta = 0 \\ 2\beta + \gamma = 0 \\ \alpha + \gamma = 0 \end{cases} \Rightarrow \begin{cases} \ominus \Rightarrow 5\alpha = 0 \Rightarrow \alpha = 0 \Rightarrow \gamma = 0 \Rightarrow \\ \Rightarrow \beta = 0 \Rightarrow \delta = 0 \\ \text{(Adevărat)} \end{cases}$$

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$\Rightarrow \{P_1, P_2, P_3, P_4\}$ sli. (1)

$\{P_1, P_2, P_3, P_4\}$ sGpt $\mathbb{R}_3[X] \Leftrightarrow \forall P_5 \in \mathbb{R}_3[X] \; \exists a, b, c, d \in \mathbb{R}$

a.î. $P_5 = aP_1 + bP_2 + cP_3 + dP_4 \Leftrightarrow$

$$\begin{cases} a + 2b + 3c + d = m \\ 2a + b = n \\ 2b + c = s \\ a + c = t \end{cases}$$ (*) sistem compatibil, unde

$P = m + nX + sX^2 + tX^3 \in \mathbb{R}_3[X]$.

$$\left(\begin{array}{cccc|c} 1 & 2 & 3 & 1 & m \\ 2 & 1 & 0 & 0 & n \\ 0 & 2 & 1 & 0 & s \\ 1 & 0 & 1 & 0 & t \end{array}\right) \xrightarrow[L_4 \to L_4 - L_1]{L_2 \to L_2 - 2L_1} \left(\begin{array}{cccc|c} 1 & 2 & 3 & 1 & m \\ 0 & -3 & -3 & -1 & n-2m \\ 0 & 2 & 1 & 0 & s \\ 0 & -2 & -2 & -1 & t-m \end{array}\right) \xrightarrow[C_3 \to C_3 - 3C_4]{C_2 \to C_2 - 3C_4}$$

$$\to \left(\begin{array}{cccc|c} 1 & -1 & 0 & 1 & m \\ 0 & 0 & 0 & -1 & n-2m \\ 0 & 2 & 1 & 0 & s \\ 0 & 1 & 1 & -1 & t-m \end{array}\right) \xrightarrow{C_2 \to C_2 - C_4} \left(\begin{array}{cccc|c} 1 & -2 & 0 & 1 & m \\ 0 & 1 & 0 & -1 & n-2m \\ 0 & 2 & 1 & 0 & s \\ 0 & 2 & 1 & -1 & t-m \end{array}\right) \to$$

$$\xrightarrow[L_4 \to L_3 - 2L_2]{L_3 \to L_3 - 2L_2} \left(\begin{array}{cccc|c} 1 & -2 & 0 & 1 & m \\ 0 & 1 & 0 & -1 & n-2m \\ 0 & 0 & 1 & 2 & s-2m+4m \\ 0 & 0 & 1 & 1 & t-2m+3m \end{array}\right) \xrightarrow{L_3 \to L_3 - L_4}$$

$$\to \left(\begin{array}{cccc|c} 1 & -2 & 0 & 1 & m \\ 0 & 1 & 0 & -1 & n-2m \\ 0 & 0 & 0 & 1 & s-t+m \\ 0 & 0 & 1 & 1 & t-2m+3m \end{array}\right) \xrightarrow{L_3 \leftrightarrow L_4} \left(\begin{array}{cccc|c} \textcircled{1} & -2 & 0 & 1 & m \\ 0 & \textcircled{1} & 0 & -1 & n-2m \\ 0 & 0 & \textcircled{1} & 1 & t-2m+3m \\ 0 & 0 & 0 & \textcircled{1} & s-t+m \end{array}\right)$$

$\Rightarrow \operatorname{rang} \begin{pmatrix} 1 & 2 & 3 & 1 \\ 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix} = 4 \Rightarrow$ sistemul (*) este comp. det. $\Rightarrow$

$\Rightarrow \{P_1, P_2, P_3, P_4\}$ sG pt $\mathbb{R}_3[X]$ (1)

(1)(2) $\Rightarrow \{P_1, P_2, P_3, P_4\}$ o bază pt $\mathbb{R}_3[X]$.

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b) Fie $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R}$ a.î. $P = \alpha_1 P_1 + \alpha_2 P_2 + \alpha_3 P_3 + \alpha_4 P_4$

$\Rightarrow 3X + X^2 = \alpha_1(1 + 2X + X^3) + \alpha_2(2 + X + 2X^2) + \alpha_3(3 + X^2 + X^3) + \alpha_4$

$\Leftrightarrow 3X + X^2 = (\alpha_1 + 2\alpha_2 + 3\alpha_3 + \alpha_4) + (2\alpha_1 + \alpha_2)X + (2\alpha_2 + \alpha_3)X^2 + (\alpha_1 + \alpha_3)X^3 \Leftrightarrow$

$$\Leftrightarrow \begin{cases} \alpha_1 + 2\alpha_2 + 3\alpha_3 + \alpha_4 = 0 \\ 2\alpha_1 + \alpha_2 = 3 \mid \cdot 2 \Rightarrow 4\alpha_1 + 2\alpha_2 = 6 \\ 2\alpha_2 + \alpha_3 = 1 \\ \alpha_1 + \alpha_3 = 0 \end{cases} \begin{matrix} \ominus \\ \Rightarrow 2\alpha_2 - \alpha_1 = 1 \end{matrix} \Rightarrow 5\alpha_1 = 5 \Rightarrow \alpha_1 = 1 \Rightarrow \alpha_3 = -1$$

$\Rightarrow \alpha_2 = 1 \Leftrightarrow \alpha_4 = 0$

$\Rightarrow$ coordonatele lui P în raport cu baza $\{P_1, P_2, P_3, P_4\}$ sunt $(1, 1, -1, 0)$.

③ Determinați o bază și dimensiunea următoarelor (sub)spații vectoriale:

a) $U = \{P \in \mathbb{R}_3[X] \mid P(1) = P(-1) ; P'(1) = P'(-1)\}$

b) $U = \left\{ \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} \in \mathbb{R}^4 \;\middle|\; \begin{cases} x - 3y + z = 0 \\ x - 2y = 0 \\ x - t = 0 \end{cases} \right\}$

Soluție:

a) Fie $P = a + bX + cX^2 + dX^3 ; a, b, c, d \in \mathbb{R}$.

$P(1) = a + b + c + d ; \; P(-1) = a - b + c - d$

$P(1) = P(-1) \Leftrightarrow a + b + c + d = a - b + c - d \Leftrightarrow b = -d$

$P'(1) = b + 2c + 3d ; \; P'(-1) = b - 2c + 3d$

$P'(1) = P'(-1) \Leftrightarrow b + 2c + 3d = b - 2c + 3d \Rightarrow c = 0$

$\Rightarrow U = \{a + bX + c \cdot X^2 + d \cdot X^3 \mid a, b, c, d \in \mathbb{R}\} =$

$= \{a + bX - bX^3 \mid a, b \in \mathbb{R}\} =$

$= \{a \cdot \underbrace{1}_{P_1} + b \underbrace{(X - X^3)}_{P_2} \mid a, b \in \mathbb{R}\} = \mathcal{L}\{P_1, P_2\}$

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$U = Sp\{P_1, P_2\} \Rightarrow \{P_1, P_2\}$ SG pt U (1)

Fie $\alpha, \beta \in \mathbb{R}$ a.î. $\alpha P_1 + \beta P_2 = 0 \Leftrightarrow$

$\Leftrightarrow \alpha \cdot 1 + \beta X - \beta X^3 = 0 \Leftrightarrow \alpha = \beta = 0 \Rightarrow$

$\rightarrow \{P_1, P_2\}$ SLi (2)

(1)(2) $\Rightarrow \{P_1, P_2\}$ bază pt U și $\dim_{\mathbb{R}} U = 2$

b) $\begin{cases} x - 3y + z = 0 \\ x - 2y = 0 \\ x - z = 0 \Rightarrow x = z \end{cases} \Rightarrow \begin{cases} 2x - 3y = 0 \\ x - 2y = 0 \mid \cdot 2 \end{cases} \Rightarrow$

$\Rightarrow \begin{cases} 2x - 3y = 0 \\ 2x - 4y = 0 \end{cases} \ominus \Rightarrow y = 0 \rightarrow x = 0 \Rightarrow z = 0$

$\Rightarrow U = \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \\ t \end{pmatrix} \in \mathbb{R}^4 \mid t \in \mathbb{R} \right\} = \left\{ t \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^4 \mid t \in \mathbb{R} \right\} =$

$= Sp\left\{ \underbrace{\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}}_{v_1} \right\} \Rightarrow \{v_1\}$ SG pt U (1)

Fie $\alpha \in \mathbb{R}$ a.î. $\alpha \cdot v_1 = 0_{\mathbb{R}^4} \Leftrightarrow \alpha = 0 \Rightarrow$

$\Rightarrow \{v_1\}$ SLi (2)

(1)(2) $\Rightarrow \{v_1\}$ bază pt U și $\dim_{\mathbb{R}} U = 1$.

④ Fie $U = \{P \in \mathbb{R}_3[X] \mid P''(0) + 2P'(0) + P(0) = 0;\ P(1) = P(-1)\}$ și $W = \{Q \in \mathbb{R}_3[X] \mid Q''(0) - 2Q'(0) + Q(0) = 0;\ Q(1) = -Q(-1)\}$ subspații ale unui spațiu vectorial V, $\dim_{\mathbb{R}} V = n$. Determinați o bază și dimensiunea subspațiilor U, W, $U \cap W$, $U + W$.

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Solutie:

$U = \{P \in \mathbb{R}_3[X] \mid P''(0) + 2P'(0) + P(0) = 0 \; ; \; P(1) = P(-1)\}$.

Fie $P = a + bX + cX^2 + dX^3 \in \mathbb{R}_3[X]$, unde $a, b, c, d \in \mathbb{R}$

$P(0) = a$; $P'(0) = b$; $P''(0) = 2c$

$\Rightarrow P''(0) + 2P'(0) + P(0) = 0 \Leftrightarrow a + 2b + 2c = 0 \Rightarrow a = -2b - 2c$

$P(1) = a + b + c + d$; $P(-1) = a - b + c - d$

$\Rightarrow P(1) = P(-1) \Leftrightarrow a + b + c + d = a - b + c - d \Leftrightarrow b = -d$

$\Rightarrow U = \{-2b - 2c + bX + cX^2 - bX^3 \mid b, c \in \mathbb{R}\} =$

$= \{b(-2 + X - X^3) + c(-2 + X^2) \mid b, c \in \mathbb{R}\} =$

$= Sp\{\underbrace{-2 + X - X^3}_{P_1}; \underbrace{-2 + X^2}_{P_2}\}$

$U = Sp\{P_1, P_2\} \Rightarrow \{P_1, P_2\}$ SG pt U (1)

Fie $\alpha_1, \alpha_2 \in \mathbb{R}$ a.î. $\alpha_1 P_1 + \alpha_2 P_2 = 0 \Leftrightarrow$

$\Leftrightarrow -2\alpha_1 + \alpha_1 X - \alpha_1 X^3 - 2\alpha_2 + \alpha_2 X^2 = 0 \Leftrightarrow \alpha_1 = \alpha_2 = 0 \Rightarrow$

$\Rightarrow \{P_1, P_2\}$ SLI (2)

(1)(2) $\Rightarrow \{P_1, P_2\}$ bază pt U și $\dim_{\mathbb{R}} U = 2$

$W = \{Q \in \mathbb{R}_3[X] \mid Q''(0) - 2Q'(0) + Q(0) = 0; Q(1) = -Q(-1)\}$

Fie $Q = m + nX + sX^2 + tX^3 \in \mathbb{R}_3[X]$, unde $m, n, s, t \in \mathbb{R}$.

$Q(0) = m$; $Q'(0) = n$; $Q''(0) = 2s$

$\Rightarrow Q''(0) - 2Q'(0) + Q(0) = 0 \Leftrightarrow 2s - 2n + m = 0 \Rightarrow m = 2n - 2s$

$Q(1) = m + n + s + t$; $Q(-1) = m - n + s - t$

$\Rightarrow Q(1) = -Q(-1) \Leftrightarrow m + n + s + t = m - n + s - t \Rightarrow n = -t \Rightarrow s = 2n$

$\Rightarrow W = \{-2n + nX + 2nX^2 + tX^3 \mid n, t \in \mathbb{R}\} =$

$= \{n(-2 + X + 2X^2) + t \cdot X^3 \mid n, t \in \mathbb{R}\} =$

$= Sp\{\underbrace{-2 + X + 2X^2}_{Q_1}; \underbrace{X^3}_{Q_2}\}$

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$W = \mathrm{Sp}\{Q_1, Q_2\} \Rightarrow \{Q_1, Q_2\}$ SG pt W (3)

Fie $\beta_1, \beta_2 \in \mathbb{R}$ a.î. $\beta_1 Q_1 + \beta_2 Q_2 = 0 \Leftrightarrow$

$\Leftrightarrow -2\beta_1 + \beta_1 X + 2\beta_1 X^2 + \beta_2 X^3 = 0 \Leftrightarrow \beta_1 = \beta_2 = 0 \rightarrow$

$\Rightarrow \{Q_1, Q_2\}$ SLi (4)

$\overset{(3)(4)}{\Longrightarrow} \{Q_1, Q_2\}$ bază pt W și $\dim_{\mathbb{R}} W = 2$.

$U \cap W = \{0\} \rightarrow \dim_{\mathbb{R}}(U \cap W) = 0$

din Th. Grassmann $\Rightarrow \dim_{\mathbb{R}}(U+W) = \dim_{\mathbb{R}} U + \dim_{\mathbb{R}} W - \dim_{\mathbb{R}}(U \cap W)$

$\Rightarrow \dim_{\mathbb{R}}(U+W) = 2+2-0 = 4$

dar $\dim_{\mathbb{R}} \mathbb{R}_3[X] = 4$ și $U+W \subseteq \mathbb{R}_3[X] \Big| \rightarrow$

$\Rightarrow U+W = \mathbb{R}_3[X]$.

(5) Determinați câte o bază și dimensiunea pentru subspațiile U, W, $U \cap W$ și $U+W$, unde $U = \mathrm{Sp}\left\{\begin{pmatrix}0&1\\1&1\end{pmatrix}, \begin{pmatrix}1&1\\1&2\end{pmatrix}, \begin{pmatrix}-2&0\\1&1\end{pmatrix}\right\}$ și $W = \mathrm{Sp}\left\{\begin{pmatrix}-1&3\\2&0\end{pmatrix}, \begin{pmatrix}1&1\\0&-1\end{pmatrix}\right\}$.

$U = \mathrm{Sp}\{\underbrace{\begin{pmatrix}0&1\\1&1\end{pmatrix}}_{u_1}, \underbrace{\begin{pmatrix}1&1\\1&2\end{pmatrix}}_{u_2}, \underbrace{\begin{pmatrix}-2&0\\1&1\end{pmatrix}}_{u_3}\} \Rightarrow \{u_1, u_2, u_3\}$ SG pt U (1)

Fie $a, b, c \in \mathbb{R}$ a.î. $au_1 + bu_2 + cu_3 = \begin{pmatrix}0&0\\0&0\end{pmatrix} \Leftrightarrow$

$\Leftrightarrow a\begin{pmatrix}0&1\\1&1\end{pmatrix} + b\begin{pmatrix}1&1\\1&2\end{pmatrix} + c\begin{pmatrix}-2&0\\1&1\end{pmatrix} = \begin{pmatrix}0&0\\0&0\end{pmatrix} \Leftrightarrow \begin{pmatrix}b-2c & a+b\\ a+b+c & a+2b+c\end{pmatrix} = \begin{pmatrix}0&0\\0&0\end{pmatrix} \Leftrightarrow$

$\Leftrightarrow \begin{cases} b-2c=0 \\ a+b=0 \\ a+b+c=0 \\ a+2b+c=0 \end{cases} \Big| \begin{matrix} \ominus \Rightarrow c=0 \\ \ominus \Rightarrow b=0 \end{matrix} \Rightarrow a=0 \Rightarrow \{u_1, u_2, u_3\}$ SLi (2)

$\overset{(1)(2)}{\Longrightarrow} \{u_1, u_2, u_3\}$ bază pt U și $\dim_{\mathbb{R}} U = 3$.

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$W = Sp\{\underbrace{\begin{pmatrix}-1&3\\2&-1\end{pmatrix}}_{w_1}, \underbrace{\begin{pmatrix}1&1\\0&-1\end{pmatrix}}_{w_2}\} \Rightarrow \{w_1, w_2\}$ SG pt W (3)

Fie $\alpha, \beta \in \mathbb{R}$ a.î. $\alpha w_1 + \beta w_2 = \begin{pmatrix}0&0\\0&0\end{pmatrix} \Leftrightarrow$

$\Leftrightarrow \alpha\begin{pmatrix}-1&3\\2&-1\end{pmatrix} + \beta\begin{pmatrix}1&1\\0&-1\end{pmatrix} = \begin{pmatrix}0&0\\0&0\end{pmatrix} \Leftrightarrow \begin{pmatrix}-\alpha+\beta & 3\alpha+\beta\\ 2\alpha & -\alpha-\beta\end{pmatrix} = \begin{pmatrix}0&0\\0&0\end{pmatrix} \Leftrightarrow$

$\Leftrightarrow \begin{cases} -\alpha+\beta = 0 \\ 3\alpha+\beta = 0 \\ 2\alpha = 0 \Rightarrow \alpha = 0 \\ -\alpha-\beta = 0 \end{cases} \Rightarrow \beta = 0 \Rightarrow \{w_1, w_2\}$ SLi (4)

(3)(4) $\{w_1, w_2\}$ bază pt W și $\dim_{\mathbb{R}} W = 2$.

Cunoaștem că $U, W \subseteq \mathcal{M}_2(\mathbb{R})$ și $B_{\mathcal{M}_2(\mathbb{R})} = \{e_1 = \begin{pmatrix}1&0\\0&0\end{pmatrix}; e_2 = \begin{pmatrix}0&1\\0&0\end{pmatrix}; e_3 = \begin{pmatrix}0&0\\1&0\end{pmatrix}; e_4 = \begin{pmatrix}0&0\\0&1\end{pmatrix}\}$

Fie A matricea care are pe coloane coordonatele vectorilor u_1, u_2, u_3, w_1, w_2 în baza canonică a lui $\mathcal{M}_2(\mathbb{R})$

$u_1 = \begin{pmatrix}0&1\\1&1\end{pmatrix} = 0\cdot e_1 + 1\cdot e_2 + 1\cdot e_3 + 1\cdot e_4$

$u_2 = \begin{pmatrix}1&1\\1&2\end{pmatrix} = 1\cdot e_1 + 1\cdot e_2 + 1\cdot e_3 + 2\cdot e_4$

$u_3 = \begin{pmatrix}-2&0\\1&1\end{pmatrix} = -2e_1 + 0\cdot e_2 + 1\cdot e_3 + 1\cdot e_4$

$w_1 = \begin{pmatrix}-1&3\\2&-1\end{pmatrix} = -e_1 + 3e_2 + 2e_3 - e_4$

$w_2 = \begin{pmatrix}1&1\\0&-1\end{pmatrix} = e_1 + e_2 + 0\cdot e_3 - e_4$

$\Rightarrow A = \begin{pmatrix}0&1&-2&-1&1\\1&1&0&3&1\\1&1&1&2&0\\1&2&1&-1&-1\end{pmatrix} \begin{matrix}\\\\\\\\ u_1\ u_2\ u_3\ w_1\ w_2\end{matrix} \xrightarrow{L_1 \leftrightarrow L_2} \begin{pmatrix}1&1&0&3&1\\0&1&-2&-1&1\\1&1&1&2&0\\1&2&1&-1&-1\end{pmatrix} \xrightarrow{L_3 \to L_3 - L_1}$

$\rightarrow \begin{pmatrix}1&1&0&3&1\\0&1&-2&-1&1\\0&0&1&-1&-1\\1&2&1&-1&-1\end{pmatrix} \xrightarrow{C_4 \to C_4 - C_5} \begin{pmatrix}1&1&0&2&1\\0&1&-2&-2&1\\0&0&1&0&-1\\1&2&1&0&-1\end{pmatrix} \xrightarrow[L_4 \to L_4 - L_3]{L_2 \to L_2 + 2L_3}$

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